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Practitioner in River and Coastal Engineering

Study location overview



Backgrounds

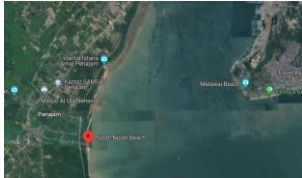
Mean coastal erosion (on average)



0.5 – 21 m/year
Luijendijk et al. (2018)



27 – 40 m/year
Studied by Maritime Affairs and Fisheries
Ministry of Indonesia



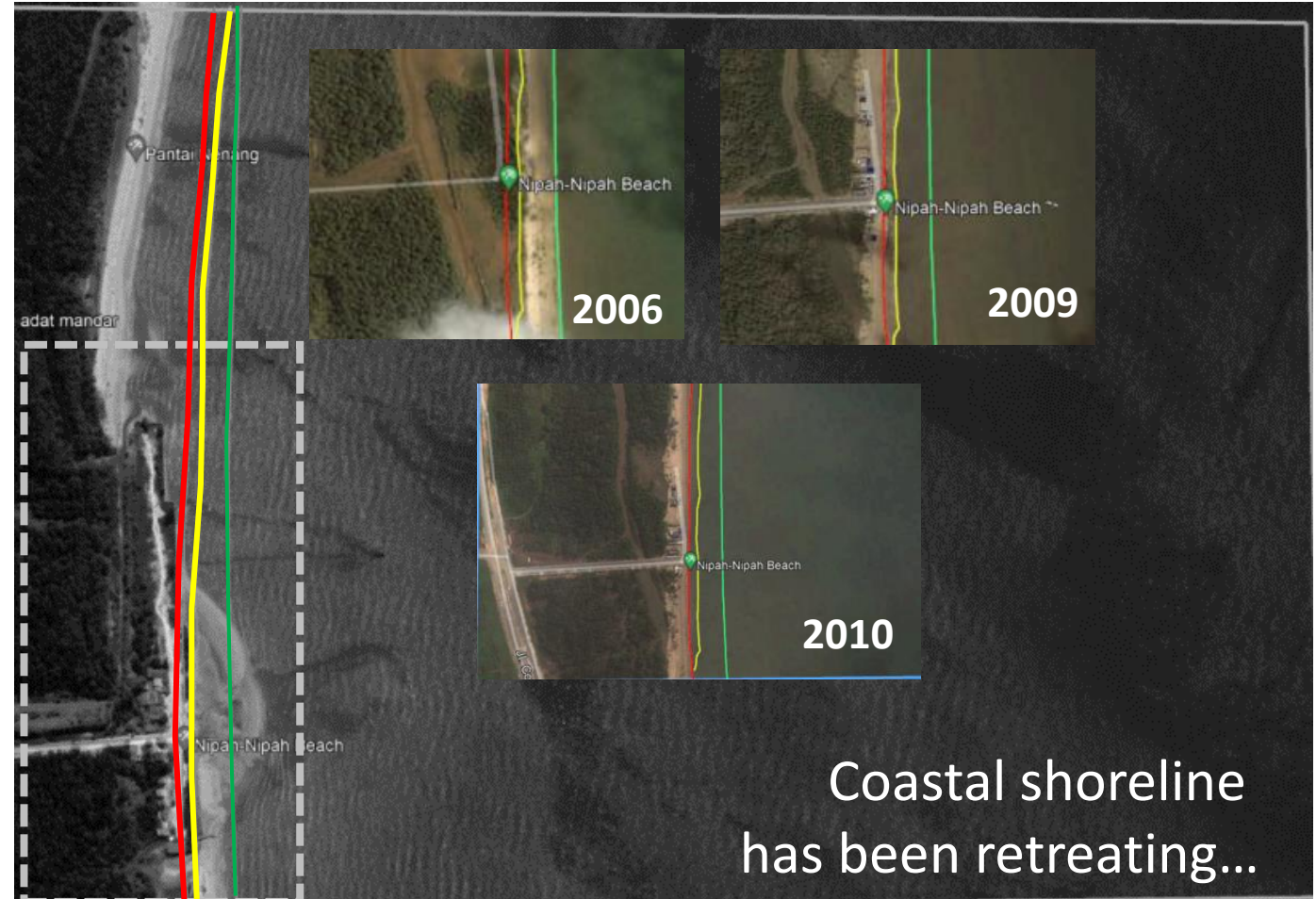
7 – 13 m/year
Approximation using Google Earth in this
study

Why important?

Population, Tourism,
Potential Hazard,
Esthetic Losses etc.

How do we solve it?

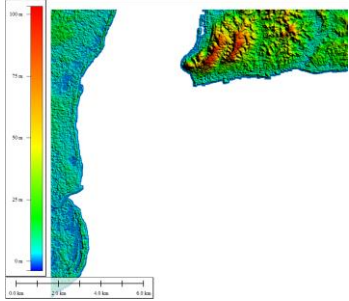
Hard and soft measures → **Vegetation**



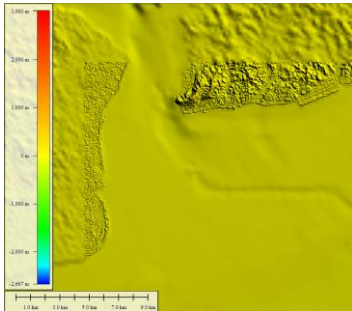
Coastal shoreline
has been retreating...

Methods

Data collection



DEM (Digital Elevation Model) - DEMNAS



Bathymetry - BATNAS

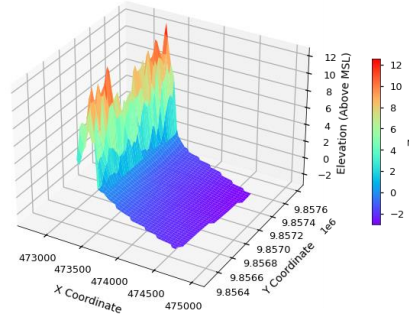


Wave condition

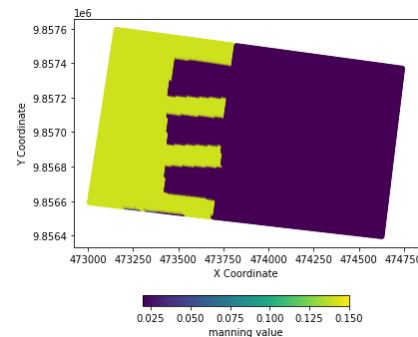


Manning values

Data and model processing in Delft3D

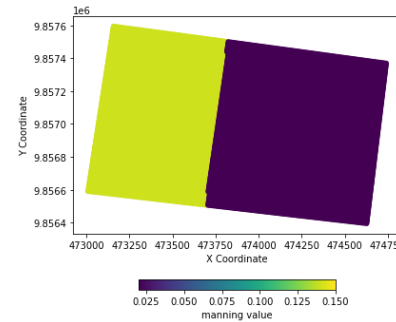


Topo-bathymetry derived from BATNAS-DEMNAS

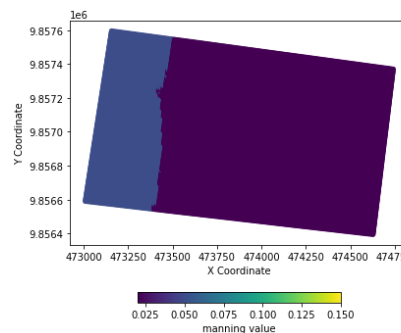


Study area with 51.7 ha of mangrove

Wave height = 0.65 m
Wave period = 3.6 s
Sediment diameter = 150



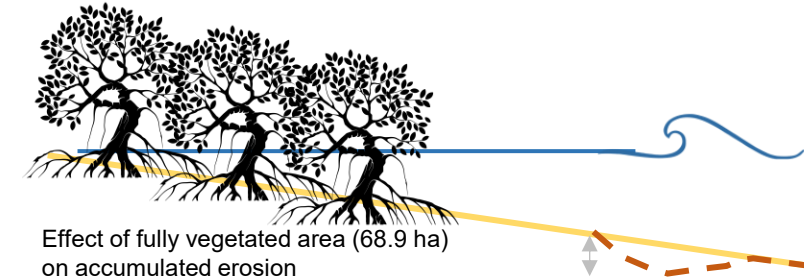
Study area with 68.9 ha of mangrove



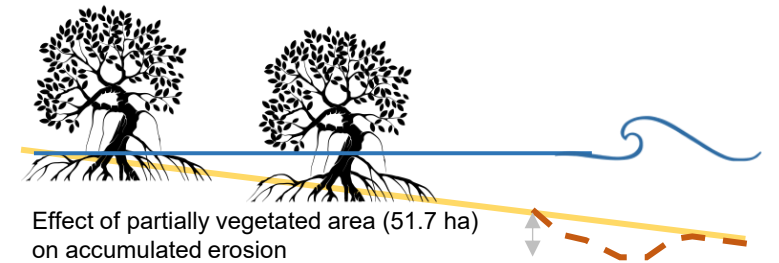
Study area without vegetation cover

HHWL (*Highest High Water Level*) = 3.33 m

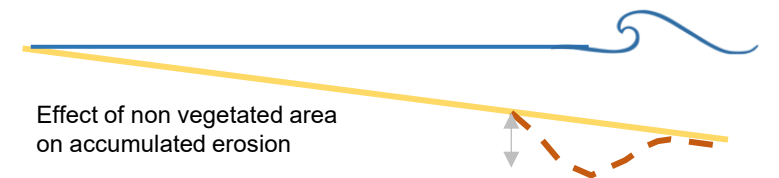
Interpretation



Effect of fully vegetated area (68.9 ha)
on accumulated erosion

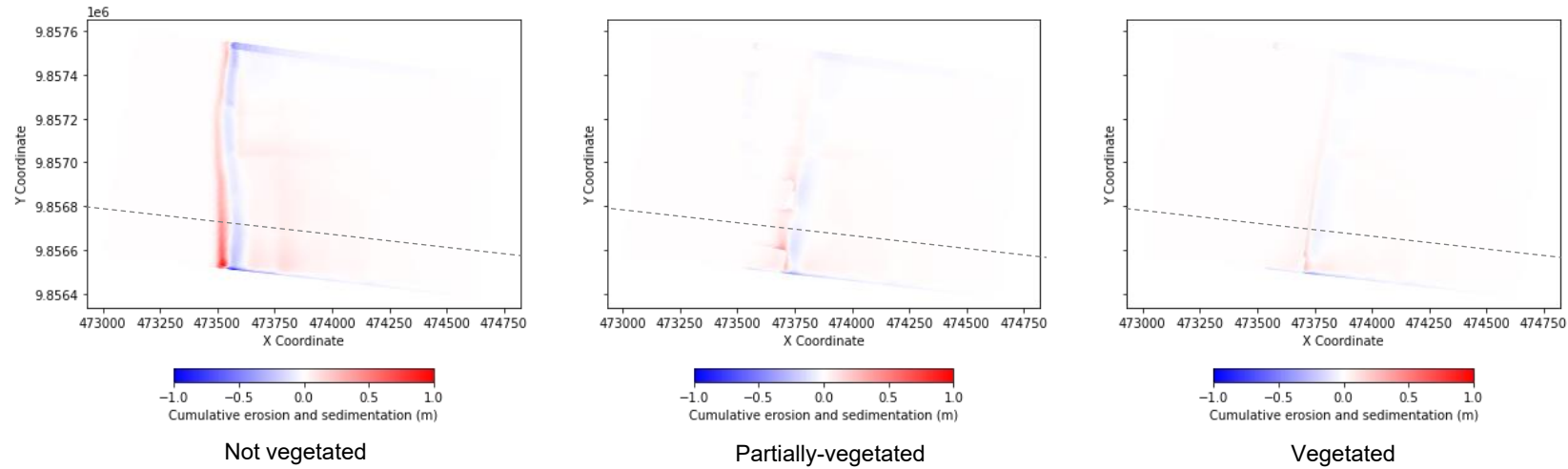


Effect of partially vegetated area (51.7 ha)
on accumulated erosion

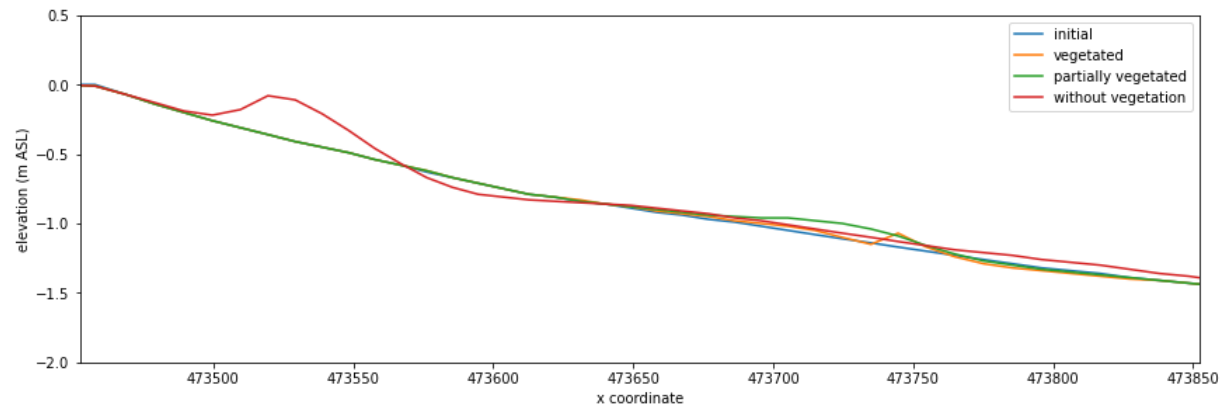


Effect of non vegetated area
on accumulated erosion

Accumulated erosion and sedimentation



A sample cross section for comparing erosion



Maximum erosion

Vegetated	0.40 m
Partially-vegetated	0.43 m
Not vegetated	0.96 m

Conclusion



1. Developed coastal area (without vegetation) could induce sediment transport, including erosion
2. The presence of vegetation (in this case: mangrove) could potentially reduce erosion by more than 50%

(assessed from maximum eroded depth and compared to the scenario without mangrove)

Main takeaway

Coastal area development should consider vegetation for ensuring coastal stability, thus preserving the coastal sustainability





Thank You

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